

Duston Moore, PhD
Independent Researcher
Edmonton, Alberta, Canada
Email: dhwcmoore@gmail.com

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Prof. Nir Piterman
Editor-in-Chief
Formal Methods in System Design
Springer

Re: Submission of Research Article: "Structural Admissibility for Verification Systems: Warrant Debt and Observational Quotients in Rocq"

Dear Prof. Piterman,

I am pleased to submit the enclosed manuscript, "Structural Admissibility for Verification Systems: Warrant Debt and Observational Quotients in Rocq," for consideration as a research article in Formal Methods in System Design.

The manuscript addresses a prior failure mode in formal verification: structural inadmissibility. I argue that some verification failures are misdiagnosed as proof-search failures when the observation interface has already collapsed distinctions required by the safety predicate. In such cases, the verification target is malformed before proof search begins.

The paper makes four main contributions. First, it formulates structural admissibility as a factorisation condition through an observational quotient. Second, it introduces warrant debt as an operational witness of representational failure, arising when algebraic observation conflates perturbations that empirical evidence distinguishes. Third, it gives a machine-checked Rocq development proving the Warrant Debt Theorem, the Non-Factorisation Theorem, and a concrete rupture decomposition theorem for composed perturbations. Fourth, it explains how these results support design-time admissibility checks and post-failure structural audit in safety-critical verification settings.

I believe the manuscript fits FMSD's scope because it concerns formal methods for the design and validation of reliable verification systems. Its central claim is that a monitor or prover must first expose enough observational structure for the target predicate to be meaningful; only then can proof search properly begin.

The accompanying Rocq development is available for review at:

<https://github.com/dhwcmoore/structural-admissibility-in-rocq>

The manuscript is original work. It has not been published elsewhere and is not currently under consideration by any other journal. I have no competing interests to disclose.

Thank you for your time and for the work you and the editorial board do for the formal methods community.

Sincerely,

Duston Moore, PhD
Independent Researcher